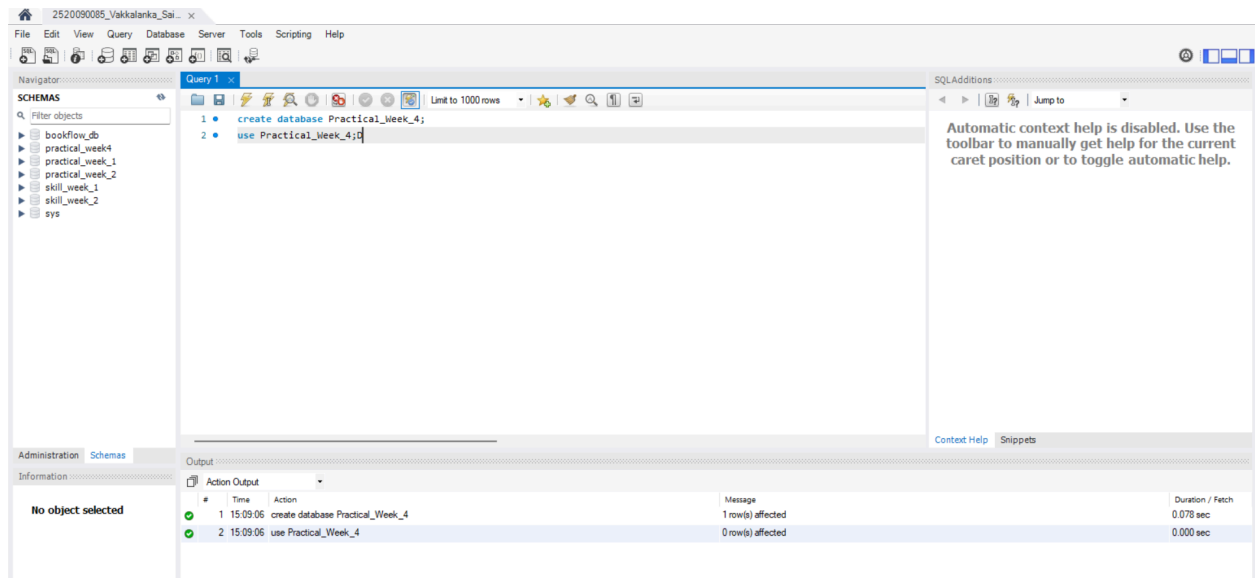


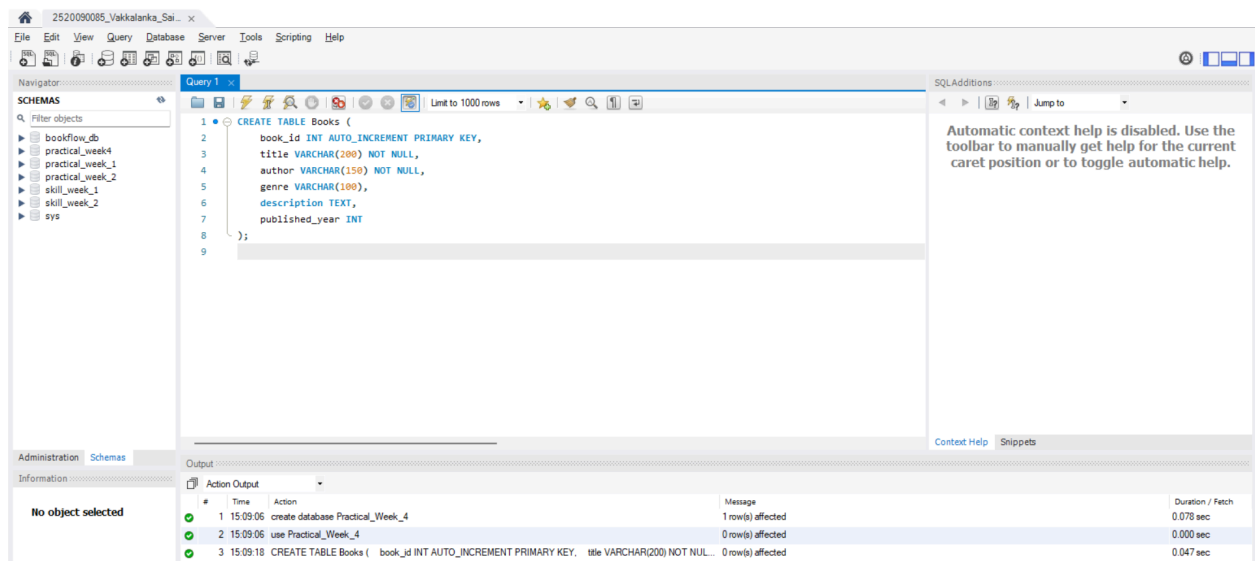
DBSE-DBD Practical Week 4

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- Roll No: 2520090085

Step 1 Create Database



Step 2 Create Books Table (DDL + Constraints)



Step 3 Describe Books Table

The screenshot shows the SQL Developer interface with the 'describe Books;' query executed. The result grid displays the table structure for the 'Books' table.

Field	Type	Null	Key	Default	Extra
book_id	int	NO	PRI	NULL	auto_increment
title	varchar(200)	NO		NULL	
author	varchar(150)	NO		NULL	
genre	varchar(100)	YES		NULL	
description	text	YES		NULL	
published_year	int	YES		NULL	

The bottom panel shows the 'Action Output' with the following details:

#	Time	Action	Message	Duration / Fetch
1	15:09:34	describe Books	6 row(s) returned	0.046 sec / 0.000 sec

Step 4 Insert 10 Books and Display

The screenshot shows the SQL Developer interface with a query that inserts 10 books and then displays them. The query is as follows:

```
7 ('Foundation', 'Isaac Asimov', 'Science Fiction', '...', 1951),
8 ('The Hobbit', 'J.R.R. Tolkien', 'Fantasy', '...', 1937),
9 ('Enders Game', 'Orson Scott Card', 'Science Fiction', '...', 1985),
10 ('Twenty Thousand Leagues Under the Seas', 'Jules Verne', 'Adventure', 'A fantastic underwater adventure aboard a myst
11 ('...', '...', '...', '...', NULL);
12
13 describe Books;
14
15 SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id;
```

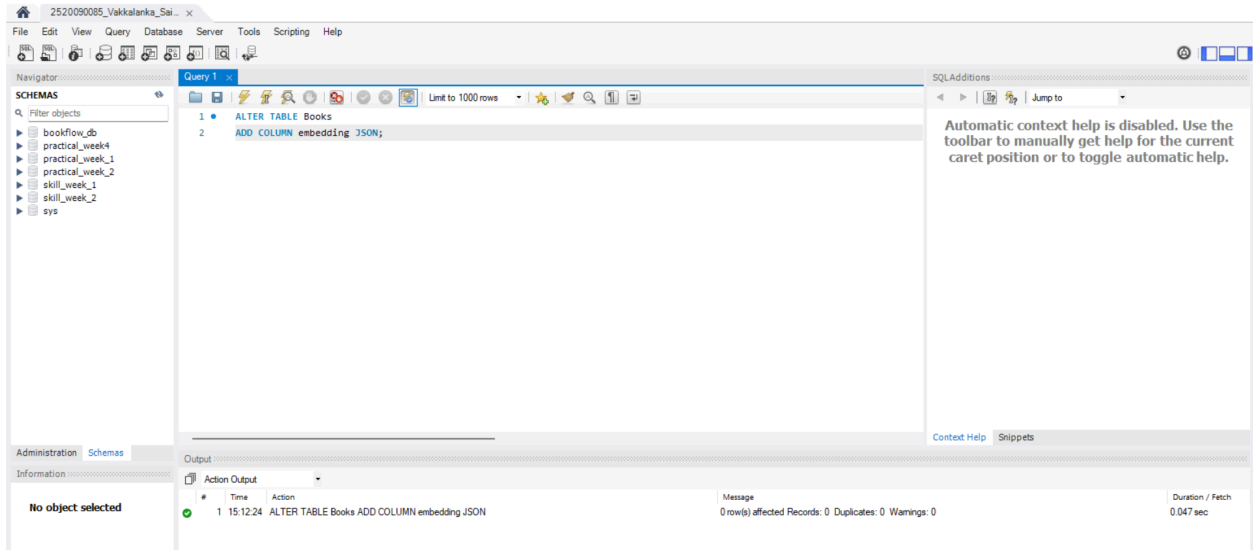
The result grid displays the following data:

book_id	title	author	genre	published_year
1	The Martian	Andy Weir	Science Fiction	2011
2	Dune	Frank Herbert	Science Fiction	1965
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	1979
4	Pride and Prejudice	Jane Austen	Romance	1813
5	The Great Gatsby	F. Scott Fitzgerald	Classic	1925
6	Foundation	Isaac Asimov	Science Fiction	1951
7	The Hobbit	J.R.R. Tolkien	Fantasy	1937
8	Enders Game	Orson Scott Card	Science Fiction	1985

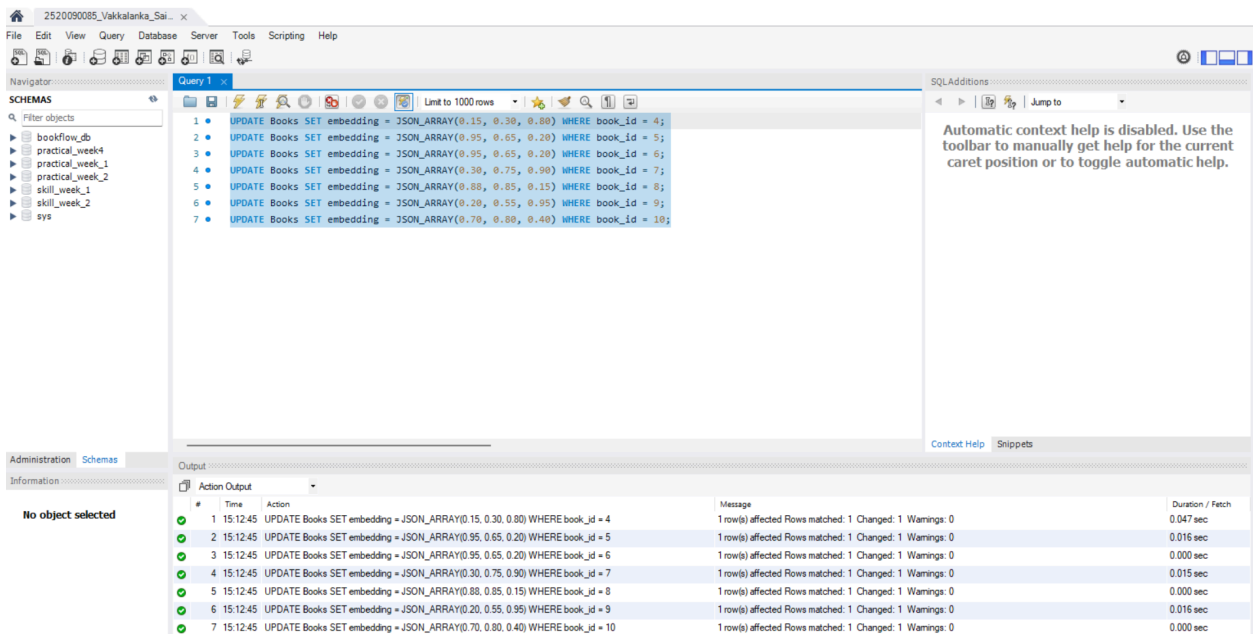
The bottom panel shows the 'Action Output' with the following details:

#	Time	Action	Message	Duration / Fetch
1	15:09:59	INSERT INTO Books (title, author, genre, description, published_year) VALUES ('The Martian', 'Andy Weir', 'Sci...	10 row(s) affected Records: 10 Duplicates: 0 Warnings: 0	0.047 sec
2	15:09:59	describe Books	6 row(s) returned	0.000 sec / 0.000 sec
3	15:09:59	SELECT book_id, title, author, genre, published_year FROM Books ORDER BY book_id LIMIT 0, 1000	10 row(s) returned	0.000 sec / 0.000 sec

Step 5 Add Embedding Column (JSON)



Step 6 Update Book Embeddings



Step 7 Display Book Embeddings

The screenshot shows the SQL Developer interface with a query window titled 'Query 1'. The query is: `SELECT book_id, title, embedding FROM Books;`. The result grid displays 8 rows of data, including book titles like 'The Martian', 'Dune', and 'The Hitchhikers Guide to the Galaxy', along with their corresponding embedding vectors. The bottom status bar indicates '10 row(s) returned'.

book_id	title	embedding
1	The Martian	[0.15, 0.30, 0.80]
2	Dune	[0.95, 0.65, 0.20]
3	The Hitchhikers Guide to the Galaxy	[0.95, 0.65, 0.20]
4	Pride and Prejudice	[0.30, 0.75, 0.90]
5	The Great Gatsby	[0.88, 0.85, 0.15]
6	Foundation	[0.88, 0.85, 0.15]
7	The Hobbit	[0.88, 0.85, 0.15]
8	Enders Game	[0.88, 0.85, 0.15]

Step 8 L2 Distance Query

The screenshot shows the SQL Developer interface with a query window titled 'Query 1'. The query is: `SELECT book_id, title, author, genre, ROUND(SQRT(POW(CAST(JSON_EXTRACT(embedding, '$[0]') AS DECIMAL(10,6)) - 0.90, 2) + POW(CAST(JSON_EXTRACT(embedding, '$[1]') AS DECIMAL(10,6)) - 0.80, 2) + POW(CAST(JSON_EXTRACT(embedding, '$[2]') AS DECIMAL(10,6)) - 0.20, 2)) AS l2_distance FROM Books;`. The result grid displays 8 rows of data, including book titles, authors, genres, and their calculated L2 distance values. The bottom status bar indicates '10 row(s) returned'.

book_id	title	author	genre	l2_distance
1	The Martian	Andy Weir	Science Fiction	0.0000
2	Dune	Frank Herbert	Science Fiction	0.0000
3	The Hitchhikers Guide to the Galaxy	Douglas Adams	Science Fiction	0.0000
4	Pride and Prejudice	Jane Austen	Romance	1.0828
5	The Great Gatsby	F. Scott Fitzgerald	Classic	0.1581
6	Foundation	Isaac Asimov	Science Fiction	0.1581
7	The Hobbit	J.R.R. Tolkien	Fantasy	0.9233
8	Enders Game	Orson Scott Card	Science Fiction	0.0735

Step 9 Cosine Similarity Query

